

EEE 306 (January 2025)

Power System I Laboratory

Final Project Report

Section: A1 Group: 02

Stability Analysis of IEEE 30 Bus System with HVDC Integration

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1. Abstract

This project investigates the stability of the IEEE 30 Bus System following the integration of High Voltage Direct Current (HVDC) transmission, using PSSE for modeling and simulation. The baseline system consists of 30 buses, 6 generators, 24 loads, 41 lines, and 4 transformers. Initial tests involved load flow analysis using the Newton-Raphson method and checking voltage and line loading limits. HVDC links were then introduced between key buses, such as Bus 2–6 and Bus 2–4, with parameters like firing angles, DC resistance, and voltage levels ranging from 250 kV to 1100 kV. Four case studies were created by varying the HVDC setup.

Simulation results revealed issues such as under voltages, excessive reactive power absorption, and power rerouting that caused line overloading. To address these, countermeasures like Static VAR Compensators (SVCs) and parallel line additions were implemented, which effectively resolved undervoltage and overloading problems. Harmonic analysis also highlighted waveform distortion, prompting recommendations for harmonic filters and equipment relocation.

Overall, the study demonstrated that HVDC integration, while initially destabilizing, can be successfully implemented with appropriate support mechanisms. It enhances transmission capability and system flexibility, offering long-term performance and economic benefits when integrated with proper control and compensation strategies.

2. Introduction

2.1 Background of Power System Stability

Power system stability is a fundamental requirement for the secure and continuous operation of power networks. It refers to the system's ability to return to steady-state conditions following a disturbance such as sudden load changes, line faults, or equipment switching. There are several types of stability — including rotor angle, voltage, and frequency stability — all of which must be carefully analyzed in modern grids, especially those integrating power electronics and renewable sources. With increasing system complexity and load demand, ensuring dynamic and steady-state stability has become more critical than ever.

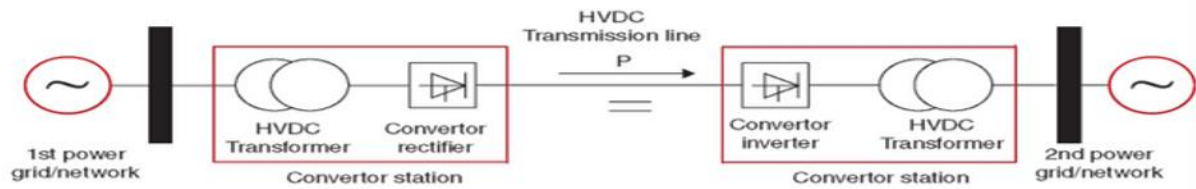
2.2 IEEE 30 Bus System Description

The IEEE 30 Bus test system is a well-established benchmark used in power system analysis for testing and validating simulation models and algorithms. It consists of 30 buses connected through 41 transmission lines and supported by 6 generators, 24 loads, and 4 transformers. The system represents a simplified portion of an actual grid (based on the American Electric Power System as of 1961) and includes sufficient complexity to study power flow, voltage control, and generator dispatch under realistic operating conditions. Its widespread use in

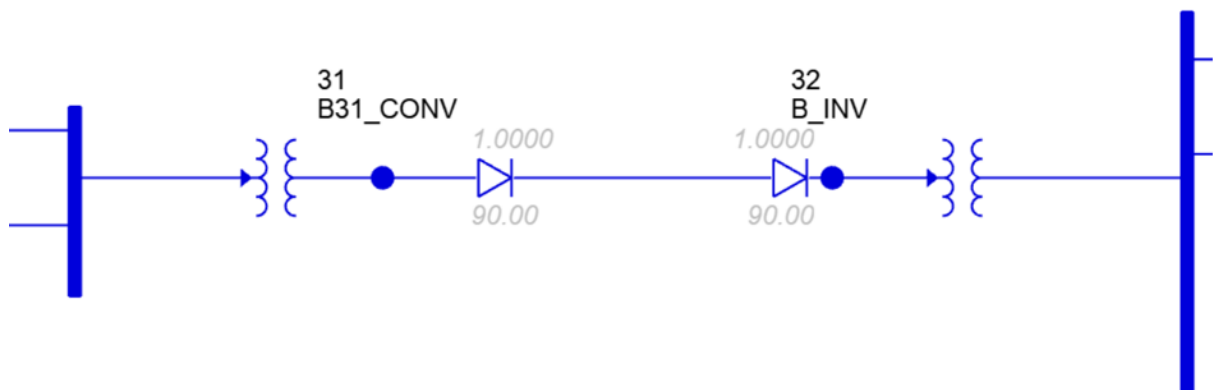
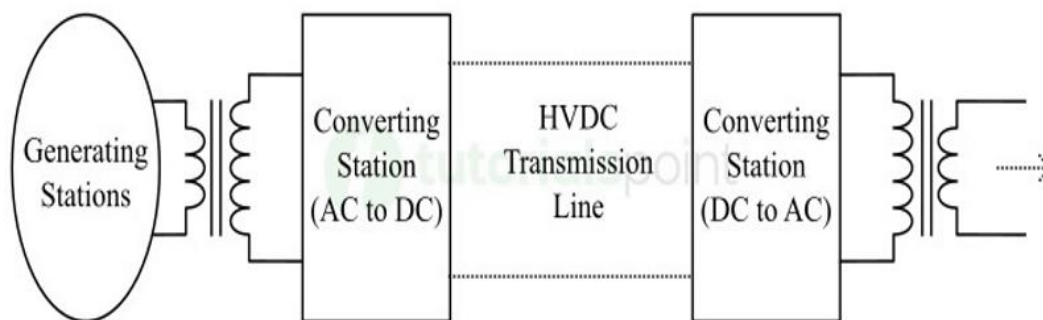
academic and industrial studies ensures consistency and comparability of simulation results.

2.3 HVDC Technology Overview

High Voltage Direct Current (HVDC) transmission systems are increasingly employed to enhance the efficiency and controllability of modern power networks.



HVDC Substation Layout



HVDC systems use converter stations to convert alternating current (AC) into direct current (DC) for transmission, and then back into AC at the receiving end. This approach offers lower transmission losses over long distances, supports asynchronous interconnection between grids, and enables precise power flow control. HVDC is especially valuable in scenarios where HVAC systems are limited by stability constraints or reactive power requirements.

2.4 Importance of HVDC Integration

Integrating HVDC into conventional AC networks can significantly improve grid reliability, stability, and power delivery efficiency. It allows operators to reroute power during congestion, regulate inter-area flows, and connect remote renewable energy sources.

However, HVDC integration also brings challenges, such as voltage fluctuations, harmonic distortions, and reactive power absorption by converters. These interactions can impact system stability if not properly addressed, making it essential to evaluate the effects of HVDC on AC grids through detailed modeling and simulation.

2.5 Project Objectives

The goal of this project is to analyze the effect of HVDC integration on the dynamic and steady-state performance of the IEEE 30 Bus system. Specific objectives include:

- Constructing an accurate simulation model of the IEEE 30 Bus network using PSSE.
- Integrating HVDC transmission links at strategic buses to simulate real-world interconnection scenarios.
- Performing load flow and stability analysis under different HVDC operating conditions.
- Identifying voltage instability, line overloading, and generator constraint violations caused by HVDC influence.
- Proposing and simulating practical countermeasures such as reactive power compensation, parallel line additions, and filter designs to ensure system reliability.

3. Literature Review

3.1 Stability Studies on IEEE Bus Systems

Numerous researchers have used IEEE bus systems, including the 14, 30, 57, and 118-bus networks, as testbeds for analyzing power system stability. These test systems offer a controlled environment for evaluating different types of disturbances and control strategies. Studies have explored topics such as voltage collapse under heavy loading, dynamic behavior during faults, and optimal placement of compensators. The IEEE 30 Bus system, in particular, has been used for testing advanced control algorithms, FACTS integration, and HVDC-based power flow control.

3.2 Previous Research on HVDC Integration

Prior research has highlighted both the benefits and complexities of integrating HVDC into AC grids. Many studies show that HVDC links can help stabilize weak AC systems, improve damping of oscillations, and enable real-time power modulation. However, other works point out the challenges, such as converter-induced harmonics, commutation failures, and the need for precise control of reactive power. Hybrid systems combining AC and DC lines require coordinated protection and control mechanisms, as well as new tools for

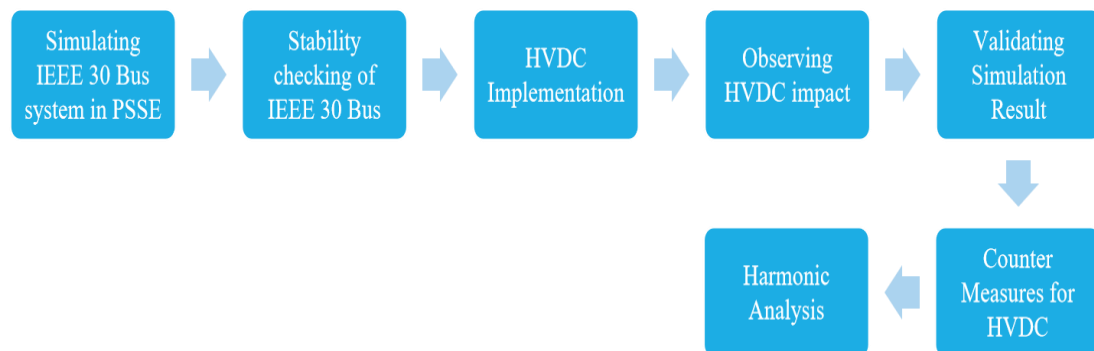
dynamic modeling and analysis.

3.3 Comparative Studies of HVDC and HVAC

Comparative analysis between HVAC and HVDC has been a central topic in transmission planning. HVDC systems have been shown to be more cost-effective for long-distance and bulk power transmission due to reduced I^2R losses and no need for reactive power support along the line. In contrast, HVAC systems are simpler to operate and maintain for short distances. Studies conclude that the decision to use HVDC over HVAC depends on multiple factors, including distance, power capacity, grid configuration, and system stability requirements.

4. Methodology

4.1 Methodology Diagram



4.2 Software Tool Description

The simulations were conducted using **PSSE (Power System Simulator for Engineering)** — an industry-standard software widely used for power flow analysis, stability studies, and network planning. PSSE provides extensive capabilities for modeling large-scale transmission systems, simulating dynamic events, and scripting custom control logic. Its user-friendly interface and comprehensive data libraries make it ideal for both academic and industrial applications.

4.3 Stability Checking Methods

4.3.1 Load Flow Analysis (Newton-Raphson Method)

The Newton-Raphson method was used for solving the power flow equations due to its fast convergence and high accuracy for large systems. It helped identify voltage drops, line overloads, and generation mismatches after integrating HVDC links.

4.3.2 Voltage Stability Checks

Voltage levels were monitored across all buses. Buses with voltage deviations exceeding $\pm 5\%$ from their nominal values were flagged. These voltage drops or surges often occurred due to reactive power absorption by the converters, indicating the need for local compensation.

4.3.3 Line Loading Limit Checks

Power flow through transmission lines was compared against their rated thermal limits. Overloading occurred near the HVDC terminals where power was rerouted through adjacent lines, especially in buses with weak grid support. These areas were targeted for reinforcement.

4.3.4 Generation Limit Checks

Generators were monitored to ensure their outputs stayed within allowable real and reactive power limits. In some cases, the presence of HVDC links caused certain generators to operate near or beyond their reactive limits, affecting voltage control and system stability.

5. IEEE 30 Bus System

5.1 IEEE 30 Bus System Modeling

5.1.1 System Configuration and Specifications

The IEEE 30 Bus system was constructed in PSSE with detailed parameters for all components including generators (with voltage setpoints and MVAR limits), transmission

lines (with impedance and thermal limits), transformers (with tap settings), and loads. The model included realistic line flows, voltage ratings, and generator characteristics, allowing accurate replication of system behavior under nominal and disturbed conditions.

5.1.2 Simulation Techniques and Procedures

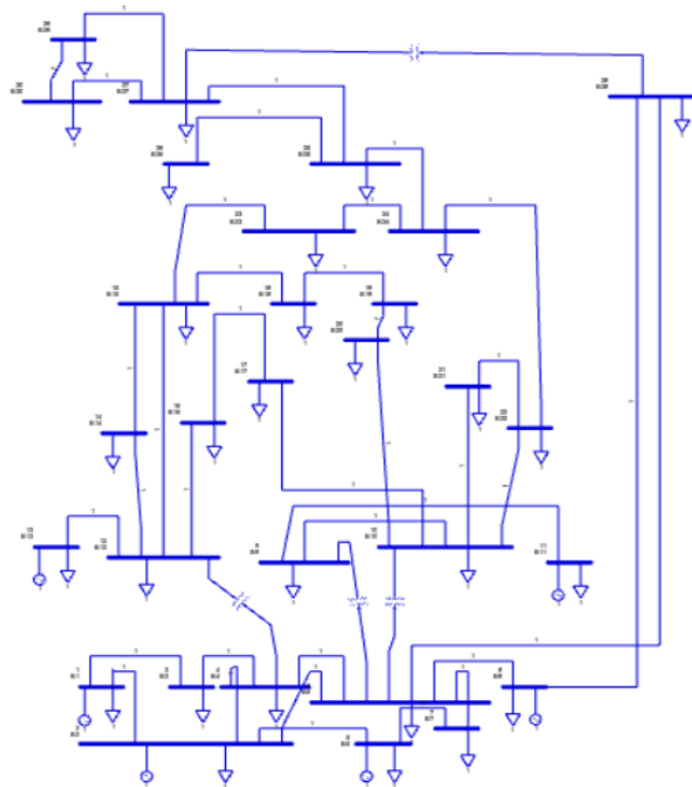


Fig: Single Line Diagram of IEEE 30 Bus System

The simulation process began with a steady-state load flow analysis to establish a baseline operating condition. Then, HVDC elements were introduced progressively in different configurations, and their impact on voltage profiles, power flows, and system limits was observed. Various operating cases were developed to simulate real-world disturbances, such as varying the converter firing angles, DC line resistance, or system loading. Each case was validated by checking convergence, voltage limits, and thermal constraints.

5.2 Simulation Results Analysis

In order to evaluate the impact of High Voltage Direct Current (HVDC) integration into the IEEE 30 Bus system, the performance of the system under base conditions without any HVDC components was analyzed. This base case serves as the benchmark for subsequent comparative studies and stability assessments.

Load Flow Analysis

The load flow analysis was conducted using the Newton-Raphson method. The simulation provided the voltage magnitudes and phase angles at each bus, along with the real and reactive power flows across all transmission elements.

Bus Number	Name	Base KV	Id	P (MW)	Q (Mvar)	QMAX (Mvar)	QMIN (Mvar)	ETERM (pu)
1	B1	424.00	1	99.31	-24.87	250.00	-20.00	1.0000
2	B2	418.00	1	80.00	10.44	100.00	-20.00	1.0000
5	B5	404.00	1	50.00	42.08	80.00	-15.00	1.0000
8	B8	404.00	1	20.00	59.28	60.00	-15.00	1.0000
11	B11	432.80	1	20.00	7.07	50.00	-10.00	1.0000
13	B13	428.40	1	20.00	13.09	60.00	-15.00	1.0000

System Stability

Observation of the contour plot shows that almost the entire region is blue, indicating stability throughout the system with no undervoltage or overloading situations.

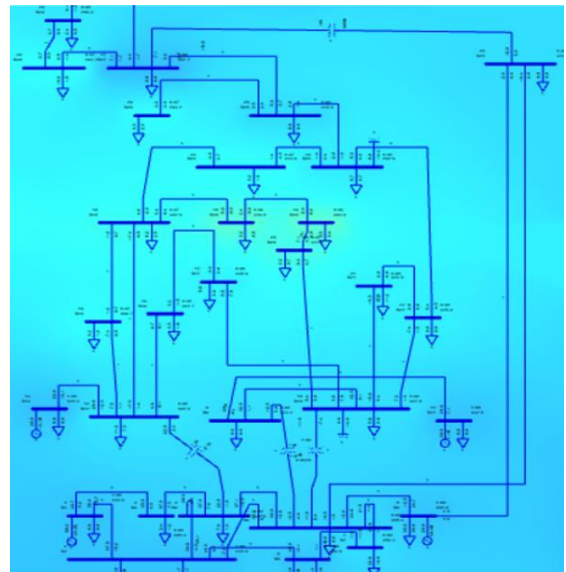
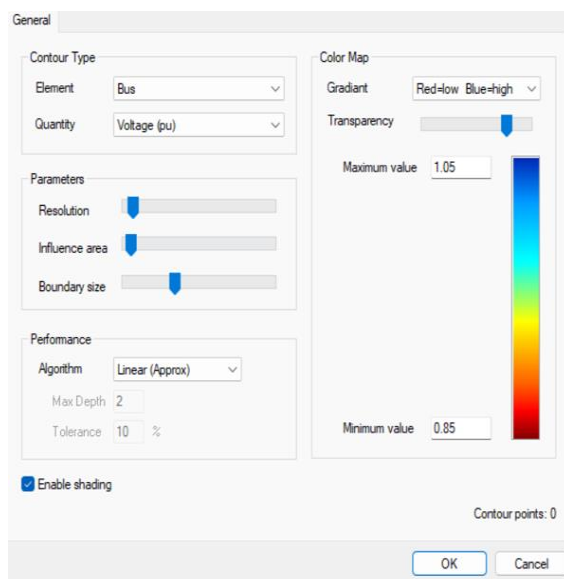


Fig: Contour plot of the IEEE 30 Bus Simulation Result

6. HVDC Integration

6.1 HVDC Model Parameter Selection

6.1.1 HVDC Line Positioning

The optimal position for HVDC line in IEEE 30 bus system is in between bus 2 to 6. So few iterations were done to pick the best suited position for the bus system used.

HVDC Line between BUS2 & BUS6:

Reached tolerance in 6 iterations

Largest mismatch: 0.05 MW 0.00 Mvar 0.05 MVA at bus 2 [B2 418.00]
System total absolute mismatch: 0.07 MVA

SWING BUS SUMMARY:

BUS#-SCT	X--	NAME	--X	BASKV	PGEN	PMAX	PMIN	QGEN	QMAX	QMIN
1	B1	424.00			185.7	200.0	50.0	-34.0*	250.0	-20.0

HVDC Line between BUS2 & BUS4:

Reached tolerance in 6 iterations

Largest mismatch: 0.03 MW 0.00 Mvar 0.03 MVA at bus 2 [B2 418.00]
System total absolute mismatch: 0.05 MVA

SWING BUS SUMMARY:

BUS#-SCT	X--	NAME	--X	BASKV	PGEN	PMAX	PMIN	QGEN	QMAX	QMIN
1	B1	424.00			189.0	200.0	50.0	-33.6*	250.0	-20.0

HVDC Line between BUS2 & BUS3:

Reached tolerance in 6 iterations

Largest mismatch: 0.03 MW -0.00 Mvar 0.03 MVA at bus 2 [B2 418.00]
System total absolute mismatch: 0.05 MVA

SWING BUS SUMMARY:

BUS#-SCT	X--	NAME	--X	BASKV	PGEN	PMAX	PMIN	QGEN	QMAX	QMIN
1	B1	424.00			187.5	200.0	50.0	-47.5*	250.0	-20.0

HVDC Line between BUS2 & BUS5:

Blown up after 3 iterations

Largest mismatch: 1605.91 MW 1792.97 Mvar 2407.00 MVA at bus 4 [B4 396.42]
System total absolute mismatch: 6054.76 MVA

SWING BUS SUMMARY:

BUS#-SCT	X--	NAME	--X	BASKV	PGEN	PMAX	PMIN	QGEN	QMAX	QMIN
1	B1	424.00			508.4*	200.0	50.0	596.8*	250.0	-20.0

From the tests we found it safe to connect the HVDC line in between Bus 2 and Bus 4 in order to get the best outputs.

6.1.2 DC Transmission Voltage Levels

Three different DC voltage levels were tested — **250 kV, 500 kV, and 1100 kV** — representing various scales of HVDC systems, from regional back-to-back links to long-distance transmission. Each voltage level influences the current flow, losses, and converter design, providing insights into the effect of HVDC sizing on system behavior.

6.1.3 Converter Parameters (Firing Angles, DC Resistance, Transformer Control Mode)

Converters were modeled with adjustable firing angles and DC resistances, reflecting the dynamic nature of thyristor-based HVDC systems. Transformer settings were configured for constant power or constant voltage control modes to evaluate how converter operations influence the surrounding AC network.

The power factor is directly related to the firing angle of the thyristors and the commutation overlap angle. Therefore, it is common for commutated HVDC converters to operate with a maximum firing delay angle of 15 degrees to 20 degrees to optimize their efficiency by minimizing the reactive power demanded by the converter.

The DC resistance value was set as 12 ohm as it was suitable for the HVDC line length used.

Line Name
HVDC

☐ Metered on Rectifier

Control Mode
Power

Vschedule (kV)
0

Setval (MW)
0

Rdcm-Ohm (ohms)
12.000

Rdc-Ohm (ohms)
12.000

Delti (pu)
0.0000

Description	Parameter	Value
	Rated DC current	2 kA
	Length of DC line	1000 km
	Resistance of DC line	0.04633 Ω/km

Resistance of DC line was 0.04633 ohm/km and the line length is 278 km as used and so the resistance value stands to be 12 ohm.

6.2 Post-HVDC Integration Analysis

Following the baseline assessment, the IEEE 30 Bus system was modified to include HVDC transmission to evaluate the impact on system performance. Four distinct test cases were developed, each incorporating different HVDC voltage levels and numbers of series-connected bridges. These cases reflect increasing levels of HVDC capacity and complexity.

The HVDC line was positioned between Bus 2 and Bus 4. Each case was simulated using PSSE to observe how HVDC alters voltage profiles, line loading and overall system stability.

6.2.1 Case-1 (250 kV, 1 Bridge)

- **Configuration:** HVDC line between Bus 2 and Bus 4, operating at a DC transmission voltage level of 250 kV with 1 bridge in series.

- *Load Flow Analysis:*

Reached tolerance in 6 iterations

Largest mismatch: 0.07 MW -0.00 Mvar 0.07 MVA at bus 2 [B2 418.00]
System total absolute mismatch: 0.14 MVA

SWING BUS SUMMARY:

BUS#-SCT	X--	NAME	--X	BASKV	PGEN	PMAX	PMIN	QGEN	QMAX	QMIN
1		B1		424.00	228.2*	200.0	50.0	-47.2*	250.0	-20.0

- *Bus Voltage Profile*

PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E Xplore FRI, JUL 18 2025 1:11

BUSES WITH VOLTAGE GREATER THAN 1.0500:

BUS#-SCT	X--	NAME	--X	BASKV	AREA	V(PU)	V(KV)	BUS#-SCT	X--	NAME	--X	BASKV	AREA	V(PU)	V(KV)
* NONE *															

BUSES WITH VOLTAGE LESS THAN 0.9500:

BUS#-SCT	X--	NAME	--X	BASKV	AREA	V(PU)	V(KV)	BUS#-SCT	X--	NAME	--X	BASKV	AREA	V(PU)	V(KV)
* NONE *															

- **Line Loading and Power Flow**

SUBSYSTEM LOADING CHECK (INCLUDED: LINES; BREAKERS AND SWITCHES; TRANSFORMERS) (EXCLUDED: NONE)
LOADINGS ABOVE 100.0 % OF BRANCH RATING (MVA FOR TRANSFORMERS, CURRENT FOR NON-TRANSFORMER BRANCHES)
LOADINGS ABOVE 100.0 % OF SWITCHING DEVICE RATING

X-----	FROM BUS	-----X	X-----	TO BUS	-----X	RATING SET 1	RATING SET 2	RATING SET 3		
BUS#-SCT	X--	NAME	--X	BASKV	AREA	CTK	LOADING	RATE1 PERCENT	RATE2 PERCENT	RATE3 PERCENT
* NONE *										

- **Observations:**

The overall system remained largely stable. The swing bus active and reactive powers changed slightly. No undervoltage conditions or line overloading occurred. The impact on the system was minimal, indicating that the network could handle this level of HVDC integration without major issues.

6.2.2 Case-2 (500 kV, 2 Bridges)

- **Configuration:** Voltage increased to 500 kV with 2 bridges in series.
- *Load Flow Analysis:*

Largest mismatch: 0.03 MW 0.00 Mvar 0.03 MVA at bus 2 [B2 418.00]
System total absolute mismatch: 0.05 MVA

SWING BUS SUMMARY:

BUS#-SCT	X--	NAME	--X	BASKV	PGEN	PMAX	PMIN	QGEN	QMAX	QMIN
1		B1		424.00	197.4	200.0	50.0	-36.8*	250.0	-20.0

- *Bus Voltage Profile*

BUSES WITH VOLTAGE GREATER THAN 1.0500:

BUS#-SCT	X--	NAME	--X	BASKV	AREA	V(PU)	V(KV)
* NONE *							

BUSES WITH VOLTAGE LESS THAN 0.9500:

BUS#-SCT	X--	NAME	--X	BASKV	AREA	V(PU)	V(KV)
18		B18		329.46	1	0.9442	311.09
20		B20		328.23	1	0.9431	309.56
22		B22		333.28	1	0.9461	315.31
24		B24		298.65	1	0.9392	280.51
30		B30		269.89	1	0.9480	255.85

- **Line Loading and Power Flow**

PSS(R)E Xplore-36.2.0 LIMIT CHECKING REPORT - BRANCH FLOWS										FRI, JUL 18 2025 1:18							
SUBSYSTEM LOADING CHECK (INCLUDED: LINES; BREAKERS AND SWITCHES; TRANSFORMERS) (EXCLUDED: NONE)																	
LOADINGS ABOVE 100.0 % OF BRANCH RATING (MVA FOR TRANSFORMERS, CURRENT FOR NON-TRANSFORMER BRANCHES)																	
LOADINGS ABOVE 100.0 % OF SWITCHING DEVICE RATING																	
X----- FROM BUS -----X X----- TO BUS -----X																	
BUS#-SCT	X--	NAME	--X	BASKV	AREA	BUS#-SCT	X--	NAME	--X	BASKV	AREA	CT	LOADING	RATING SET 1	RATING SET 2	RATING SET 3	
1	B1			424.00*	1	2	B2			418.00	1	1	310.1	130.0	238.5	130.0	238.5
2	B2			418.00	1	6	B6			391.79*	1	1	164.4	65.0	253.0	65.0	253.0
4	B4			396.42*	1	6	B6			391.79	1	1	317.3	90.0	352.5	90.0	352.5
5	B5			404.00	1	7	B7			391.77*	1	1	99.5	70.0	142.1	70.0	142.1

- **Observations:**

Increasing the HVDC transmission voltage led to more pronounced changes in system behavior. Voltage level of 10 buses drop below 0.95 p.u. The power injected through the HVDC line caused higher loading on adjacent transmission lines, particularly those connecting Bus 2 to downstream areas. 4 lines get overloaded with maximum loading up to 352.5 percent.

6.2.3 Case-3 (800 kV, 3 Bridges)

- **Configuration:** HVDC line operating at 800 kV with 3 bridges.
- *Load Flow Analysis:*

Largest mismatch: 0.03 MW 0.00 Mvar 0.03 MVA at bus 2 [B2 418.00]
 System total absolute mismatch: 0.06 MVA

SWING BUS SUMMARY:

BUS#-SCT	X--	NAME	--X	BASKV	PGEN	PMAX	PMIN	QGEN	QMAX	QMIN
1	B1	424.00			191.3	200.0	50.0	-34.5*	250.0	-20.0

- *Bus Voltage Profile*

BUSES WITH VOLTAGE GREATER THAN 1.0500:

BUS#-SCT	X--	NAME	--X	BASKV	AREA	V(PU)	V(KV)	BUS#-SCT	X--	NAME	--X	BASKV	AREA	V(PU)	V(KV)
* NONE *															

BUSES WITH VOLTAGE LESS THAN 0.9500:

BUS#-SCT	X--	NAME	--X	BASKV	AREA	V(PU)	V(KV)	BUS#-SCT	X--	NAME	--X	BASKV	AREA	V(PU)	V(KV)
18	B18	329.46	1	0.9440	311.02	19	B19	315.97	1	0.9394	296.83				
20	B20	328.23	1	0.9429	309.49	21	B21	332.51	1	0.9448	314.16				
22	B22	333.28	1	0.9458	315.22	23	B23	323.39	1	0.9456	305.81				
24	B24	298.65	1	0.9390	280.43	26	B26	271.43	1	0.9398	255.10				
30	B30	269.89	1	0.9477	255.76	31	B2_REC	800.00	1	0.9474	757.93				

- **Line Loading and Power Flow**

SUBSYSTEM LOADING CHECK (INCLUDED: LINES; BREAKERS AND SWITCHES; TRANSFORMERS) (EXCLUDED: NONE)
 LOADINGS ABOVE 100.0 % OF BRANCH RATING (MVA FOR TRANSFORMERS, CURRENT FOR NON-TRANSFORMER BRANCHES)
 LOADINGS ABOVE 100.0 % OF SWITCHING DEVICE RATING

X----- FROM BUS -----X X----- TO BUS -----X										RATING SET 1		RATING SET 2		RATING SET 3					
BUS#-SCT	X--	NAME	--X	BASKV	AREA	BUS#-SCT	X--	NAME	--X	BASKV	AREA	CKT	LOADING	RATE1	PERCENT	RATE2	PERCENT	RATE3	PERCENT
1	B1	424.00*	1	2	B2	418.00	1	1	305.3	130.0	234.8	130.0	234.8	130.0	234.8				
2	B2	418.00	1	6	B6	391.79*	1	1	168.1	65.0	258.7	65.0	258.7	65.0	258.7				
4	B4	396.42*	1	6	B6	391.79	1	1	322.7	90.0	358.6	90.0	358.6	90.0	358.6				
5	B5	404.00	1	7	B7	391.77*	1	1	101.4	70.0	144.8	70.0	144.8	70.0	144.8				

- **Observations:**

This system shows similar undervoltage and overloading conditions as case-2. Overloading at 4 lines and undervoltage at 10 buses were observed. The voltage levels at the buses were lower than that of case-2 and the overloading at the lines were higher than that of case-2.

6.2.4 Case-4 (1100 kV, 4 Bridges)

- **Configuration:** HVDC link raised to 1100 kV with 4 bridges in series.
- *Load Flow Analysis:*

Reached tolerance in 6 iterations

Largest mismatch: 0.04 MW 0.00 Mvar 0.04 MVA at bus 2 [B2 418.00]
System total absolute mismatch: 0.06 MVA

SWING BUS SUMMARY:

BUS#-SCT	X--	NAME	--X	BASKV	PGEN	PMAX	PMIN	QGEN	QMAX	QMIN
1		B1		424.00	189.4	200.0	50.0	-33.8*	250.0	-20.0

- *Bus Voltage Profile*

BUS#-SCT	X--	NAME	--X	BASKV	AREA	V(PU)	V(KV)	BUS#-SCT	X--	NAME	--X	BASKV	AREA	V(PU)	V(KV)
* NONE *															

BUSES WITH VOLTAGE LESS THAN 0.9500:

BUS#-SCT	X--	NAME	--X	BASKV	AREA	V(PU)	V(KV)	BUS#-SCT	X--	NAME	--X	BASKV	AREA	V(PU)	V(KV)
18		B18		329.46	1	0.9439	310.98	19		B19		315.97	1	0.9393	296.79
20		B20		328.23	1	0.9428	309.44	21		B21		332.51	1	0.9447	314.11
22		B22		333.28	1	0.9457	315.17	23		B23		323.39	1	0.9455	305.76
24		B24		298.65	1	0.9388	280.38	26		B26		271.43	1	0.9396	255.05
30		B30		269.89	1	0.9474	255.70	31		B2_REC		1100.0	1	0.9474	1042.2

- **Line Loading and Power Flow**

SUBSYSTEM LOADING CHECK (INCLUDED: LINES; BREAKERS AND SWITCHES; TRANSFORMERS) (EXCLUDED: NONE)
LOADINGS ABOVE 100.0 % OF BRANCH RATING (MVA FOR TRANSFORMERS, CURRENT FOR NON-TRANSFORMER BRANCHES)
LOADINGS ABOVE 100.0 % OF SWITCHING DEVICE RATING

X----- FROM BUS -----X X----- TO BUS -----X								RATING SET 1		RATING SET 2		RATING SET 3							
BUS#-SCT	X--	NAME	--X	BASKV	AREA	BUS#-SCT	X--	NAME	--X	BASKV	AREA	CKT	LOADING	RATE1	PERCENT	RATE2	PERCENT	RATE3	PERCENT
1		B1		424.00*	1	2		B2		418.00	1	1	306.0	130.0	235.4	130.0	235.4	130.0	235.4
2		B2		418.00	1	6		B6		391.79*	1	1	167.5	65.0	257.8	65.0	257.8	65.0	257.8
4		B4		396.42*	1	6		B6		391.79	1	1	321.9	90.0	357.6	90.0	357.6	90.0	357.6
5		B5		404.00	1	7		B7		391.77*	1	1	101.1	70.0	144.4	70.0	144.4	70.0	144.4

- **Observations:**

Here the IEEE 30 Bus system encountered its most severe operating conditions. Extensive undervoltage occurred across the 10 buses from case-2 and case-3. The same 4 lines showed overloading conditions. The system's dynamic response became unstable, signaling that HVDC integration at this scale requires substantial reinforcement and control to avoid collapse.

6.3 Undervoltage Analysis

Undervoltage issues were progressively observed as HVDC capacity increased across the test cases. The thyristor-based HVDC converters operate as constant power loads and draw significant reactive power from the system, particularly when operating at higher firing and commutation overlap angles. This increased reactive power demand, combined with limited local voltage support at distant buses, caused noticeable voltage drops in areas far from the HVDC injection point. In the 800 kV and 1100 kV cases, several load-side buses experienced voltage levels below acceptable thresholds, particularly under high-load conditions or during minor system disturbances. The system's conventional voltage regulation mechanisms, such as generator AVR controls, were insufficient to counteract the localized voltage sags introduced by HVDC.

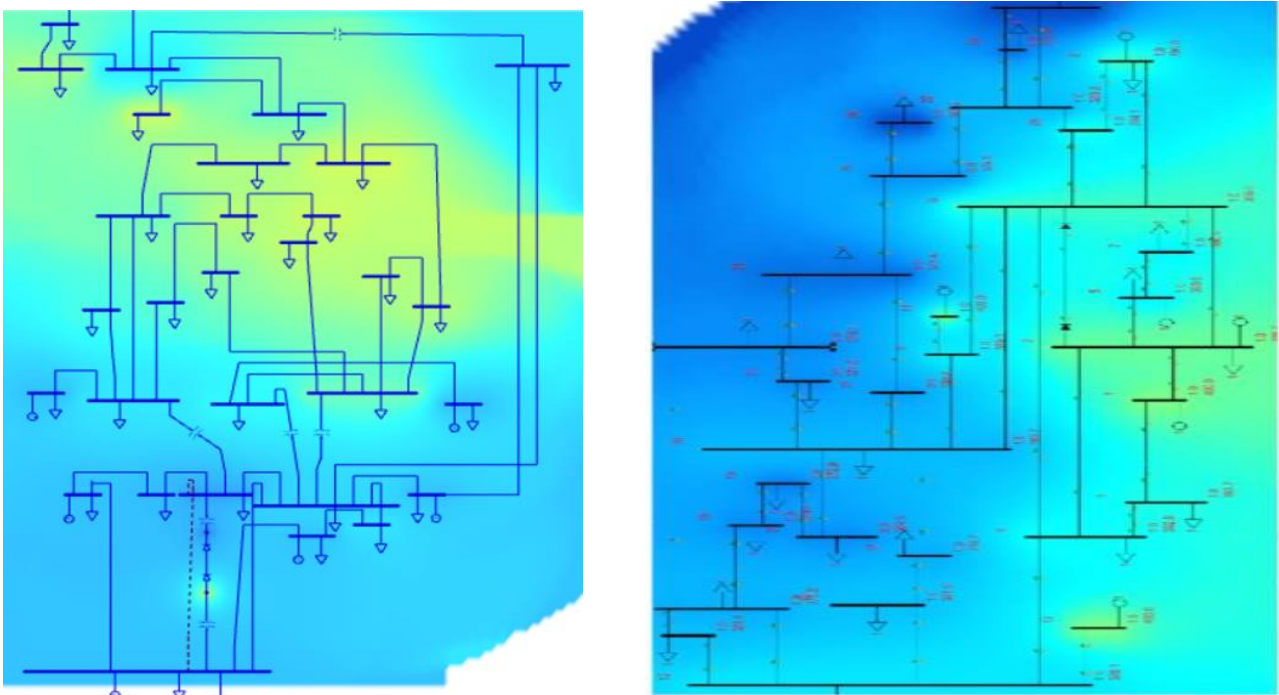


Fig: Contour Plot of the Undervoltage Situation from Our Simulation(left) vs Contour Plot of the Undervoltage Situation from a Reference Paper (right)

6.4 Line Overloading Analysis

The integration of HVDC into the IEEE 30 Bus system significantly altered the power flow distribution, resulting in overloading of several transmission lines. The HVDC link, especially when positioned near generation buses like Bus 2, rerouted large portions of the system's power through a limited number of corridors. Lines which connect critical regions of the network experienced consistently high loading in the 500 kV and higher voltage scenarios. This is attributed to the increased injection of active power at the HVDC rectifier side and concentrated withdrawal at the inverter side, effectively bypassing alternate routes.

As HVDC voltage increased, more power was transferred through fewer lines, creating bottlenecks and thermal stress.

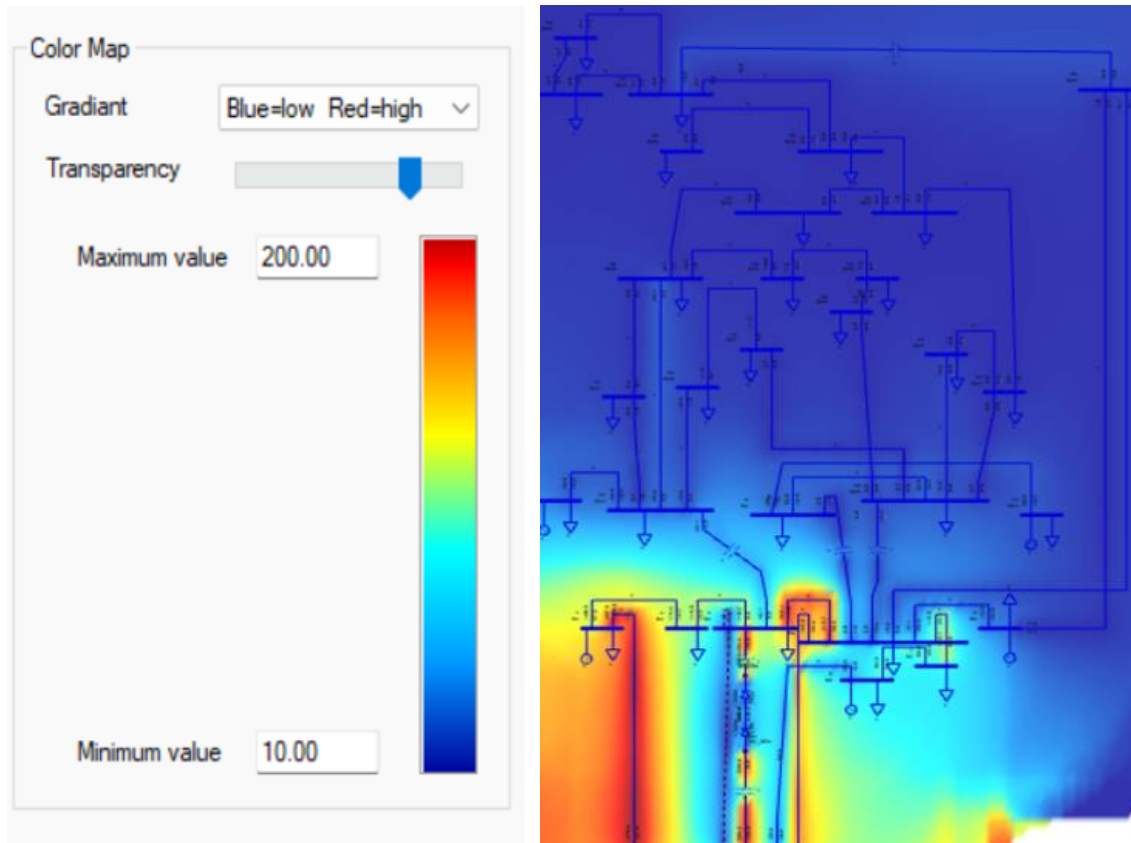


Fig: Contour Plot of the Line Overloading after HVDC Integration

The root cause lies in the reactive power demand characteristics of thyristor-based HVDC converters. These converters, commonly used in line-commutated converter (LCC) HVDC systems, draw significant amounts of reactive power from the AC network to maintain constant active power transfer.

The reactive power consumption Q of such converters can be approximated by the relationship:

$$Q \approx P \cdot \tan\left(\frac{\alpha + \mu}{2}\right)$$

where:

- P is the active power transferred
- α is the firing angle of the converter
- μ is the commutation overlap angle

This equation shows that reactive power demand increases with larger values of α and μ ,

which are typical in practical HVDC operations, especially under varying load or network conditions.

Due to this behavior, HVDC converters act as constant power loads that do not inherently support local voltage regulation. Instead, they absorb reactive power from the surrounding AC system, weakening voltage stability, particularly at buses far from generators or reactive power sources. This leads to limited voltage regulation capability, causing notable voltage drops in areas not directly supported by generator excitation systems. Furthermore, the injection and withdrawal of large amounts of power by HVDC links lead to redistribution of power flows across the network. This can cause unbalanced loading and shifting of voltage profiles, further aggravating undervoltage conditions.

7. Designing and Evaluating Counter Measures

The increasing instability observed in the IEEE 30 Bus system following HVDC integration—particularly undervoltage conditions and line overloading necessitated the design and evaluation of appropriate countermeasures. This section details the simulation-based analysis of voltage and thermal mitigation strategies and evaluates their performance in restoring system stability and reliability.

7.1 Mitigation of Undervoltage Issues

Undervoltage conditions were identified as one of the primary challenges arising from the integration of HVDC links. To counter this, two mitigation strategies were explored: the use of Static VAR Compensators (SVCs) and fixed shunt capacitors.

7.1.1 Static VAR Compensator (SVC) Analysis

Static VAR Compensators (SVCs) were introduced into the system as dynamic reactive power support devices to mitigate the undervoltage conditions caused by the reactive demand of HVDC converters. SVCs were placed at buses experiencing the largest voltage drops (bus 19 and bus 31) in high-voltage HVDC scenarios (e.g., 800 kV and 1100 kV cases).

Bus	SVC Name	MVAR Range
19	SVC1	-20 to +30 MVAR
31	SVC4	-40 to +40 MVAR

The simulation results showed a significant improvement in voltage profiles at remote load buses. SVCs dynamically injected reactive power as needed, stabilizing voltage during both steady-state and transient conditions. Their ability to adapt to system changes made them particularly effective in maintaining acceptable voltage levels under varying load and converter operating conditions.

BUSES WITH VOLTAGE GREATER THAN 1.0500:

```
BUS#-SCT X-- NAME --X BASKV AREA V(PU) V(KV) | BUS#-SCT X-- NAME --X BASKV AREA V(PU) V(KV)
* NONE *
```

BUSES WITH VOLTAGE LESS THAN 0.9500:

```
BUS#-SCT X-- NAME --X BASKV AREA V(PU) V(KV) | BUS#-SCT X-- NAME --X BASKV AREA V(PU) V(KV)
* NONE *
```

7.1.2 Fixed Shunt Capacitor Analysis

Fixed shunt capacitors were also tested as a simpler, cost-effective alternative to SVCs. These capacitors were deployed at buses that demonstrated persistent undervoltage in the post-HVDC scenarios. Bus 19, 24 and 31 were chosen for this.

Bus	Shunt Name	MVAR
19	SHUNT2	10.00
24	SHUNT3	10.00
31	SHUNT4	40.00

Although they lack dynamic control, fixed shunts provided a consistent source of reactive power, helping to improve local voltages. Simulation results confirmed that fixed capacitors successfully raised bus voltages back within acceptable limits in most cases, though not as effectively or adaptively as SVCs under changing system conditions. They remain a viable option for budget-constrained systems or as supplementary support alongside SVCs.

BUSES WITH VOLTAGE GREATER THAN 1.0500:

BUS#-SCT	X--	NAME	--X	BASKV	AREA	V(PU)	V(KV)
* NONE *							

BUSES WITH VOLTAGE LESS THAN 0.9500:

BUS#-SCT	X--	NAME	--X	BASKV	AREA	V(PU)	V(KV)
* NONE *							

7.2 Mitigation of Line Overloading Issues

7.2.1 Impact of Parallel Transmission Lines

To address the line congestion issues, most notably on lines L26 and L46, parallel transmission lines were added. These reinforcements distributed the increased power flow resulting from HVDC injection, thereby reducing thermal loading on existing corridors. Initially, one parallel line was added to each of the overloaded paths, resulting in a visible decrease in line utilization levels.

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SUBSYSTEM LOADING CHECK (INCLUDED: LINES; BREAKERS AND SWITCHES; TRANSFORMERS) (EXCLUDED: NONE)
LOADINGS ABOVE 100.0 % OF BRANCH RATING (MVA FOR TRANSFORMERS, CURRENT FOR NON-TRANSFORMER BRANCHES)
LOADINGS ABOVE 100.0 % OF SWITCHING DEVICE RATING

X----- FROM BUS -----X X----- TO BUS -----X					RATING SET 1		RATING SET 2		RATING SET 3										
BUS#-SCT	X--	NAME	--X	BASKV	AREA	BUS#-SCT	X--	NAME	--X	BASKV	AREA	CKT	LOADING	RATE1	PERCENT	RATE2	PERCENT	RATE3	PERCENT
1	B1	424.00	1	2	B2	418.00*	1	1	131.8	130.0	101.4	130.0	101.4	130.0	101.4	130.0	101.4	130.0	101.4
1	B1	424.00	1	2	B2	418.00*	1	2	131.8	130.0	101.4	130.0	101.4	130.0	101.4	130.0	101.4	130.0	101.4
2	B2	418.00	1	6	B6	391.79*	1	1	152.8	65.0	235.1	65.0	235.1	65.0	235.1	65.0	235.1	65.0	235.1
4	B4	396.42*	1	6	B6	391.79	1	1	163.0	90.0	181.1	90.0	181.1	90.0	181.1	90.0	181.1	90.0	181.1
4	B4	396.42*	1	6	B6	391.79	1	2	163.0	130.0	125.4	130.0	125.4	130.0	125.4	130.0	125.4	130.0	125.4

7.2.2 Comparative Analysis of Parallel Line Configurations

Subsequent simulations explored the effect of adding multiple parallel lines. When a second set of parallel lines was added to L26 and L46, the loading on the original lines dropped further.

SUBSYSTEM LOADING CHECK (INCLUDED: LINES; BREAKERS AND SWITCHES; TRANSFORMERS) (EXCLUDED: NONE)
LOADINGS ABOVE 100.0 % OF BRANCH RATING (MVA FOR TRANSFORMERS, CURRENT FOR NON-TRANSFORMER BRANCHES)
LOADINGS ABOVE 100.0 % OF SWITCHING DEVICE RATING

X----- FROM BUS -----X X----- TO BUS -----X					RATING SET 1		RATING SET 2		RATING SET 3										
BUS#-SCT	X--	NAME	--X	BASKV	AREA	BUS#-SCT	X--	NAME	--X	BASKV	AREA	CKT	LOADING	RATE1	PERCENT	RATE2	PERCENT	RATE3	PERCENT
2	B2	418.00	1	6	B6	391.79*	1	1	105.5	65.0	162.4	65.0	162.4	65.0	162.4	65.0	162.4	65.0	162.4
4	B4	396.42*	1	6	B6	391.79	1	1	178.0	90.0	197.8	90.0	197.8	90.0	197.8	90.0	197.8	90.0	197.8
4	B4	396.42*	1	6	B6	391.79	1	2	178.0	130.0	137.0	130.0	137.0	130.0	137.0	130.0	137.0	130.0	137.0

A total of four parallel lines, distributed across both affected paths, fully resolved the congestion issue.

SUBSYSTEM LOADING CHECK (INCLUDED: LINES; BREAKERS AND SWITCHES; TRANSFORMERS) (EXCLUDED: NONE)
 LOADINGS ABOVE 100.0 % OF BRANCH RATING (MVA FOR TRANSFORMERS, CURRENT FOR NON-TRANSFORMER BRANCHES)
 LOADINGS ABOVE 100.0 % OF SWITCHING DEVICE RATING

X----- FROM BUS -----X X----- TO BUS -----X			RATING SET 1		RATING SET 2		RATING SET 3			
BUS#-SCT X--	NAME --X	BASKV AREA	BUS#-SCT X--	NAME --X	BASKV AREA	CKT	LOADING	RATE1 PERCENT	RATE2 PERCENT	RATE3 PERCENT
* NONE *										

Power flow became more balanced throughout the network, and the system was able to absorb high-capacity HVDC power without stressing individual corridors. This analysis confirms the effectiveness of line reinforcement as a robust solution for HVDC-induced congestion.

7.3 Performance Evaluation of Counter Measures

7.3.1 Rechecking Bus Voltages and Line Loading

Following the implementation of both reactive compensation and transmission reinforcement, the system was re-simulated to assess its post-mitigation performance. Voltage levels across all buses were brought within the acceptable range, even in the high-voltage HVDC scenarios. Previously overloaded lines were found to be operating well within their thermal capacity.

Reached tolerance in 5 iterations

Largest mismatch:	-0.09 MW	-0.00 Mvar	0.09 MVA at bus 2 [B2	418.00]
System total absolute mismatch:			0.18 MVA	

SWING BUS SUMMARY:

BUS#-SCT X--	NAME --X	BASKV	PGEN	PMAX	PMIN	QGEN	QMAX	QMIN
1	B1	424.00	124.4	200.0	50.0	-1.2	250.0	-20.0

7.3.2 Generation Dispatch Stability

Generation dispatch was also reviewed in the mitigated system. The presence of SVCs and improved line capacity allowed generators to operate closer to their optimal setpoints, reducing unnecessary reactive support burden and enhancing economic operation. No instability or abnormal dispatch behavior was observed during the simulation.

Bus Number	Name	Base KV	Id	P (MW)	Q (Mvar)	QMAX (Mvar)	QMIN (Mvar)
1	B1	424.00	1	124.40	-1.22	250.00	-20.00
2	B2	418.00	1	80.00	100.00	100.00	-20.00
5	B5	404.00	1	50.00	50.61	80.00	-15.00
8	B8	404.00	1	20.00	26.15	60.00	-15.00
11	B11	432.80	1	20.00	5.79	50.00	-10.00
13	B13	428.40	1	20.00	6.82	60.00	-15.00
19	B19	315.97	1	0.00	22.49	30.00	-20.00
31	B2_REC	500.00	1	0.00	40.00	40.00	-40.00

7.4 Harmonic Analysis

After implementing the countermeasures to resolve undervoltage and line overloading issues, a harmonic distortion analysis was conducted to evaluate the overall power quality of the HVDC-integrated IEEE 30 Bus system. This step was critical, as HVDC systems are known sources of harmonic currents which can propagate through the network and affect sensitive equipment, voltage waveforms, and system stability.

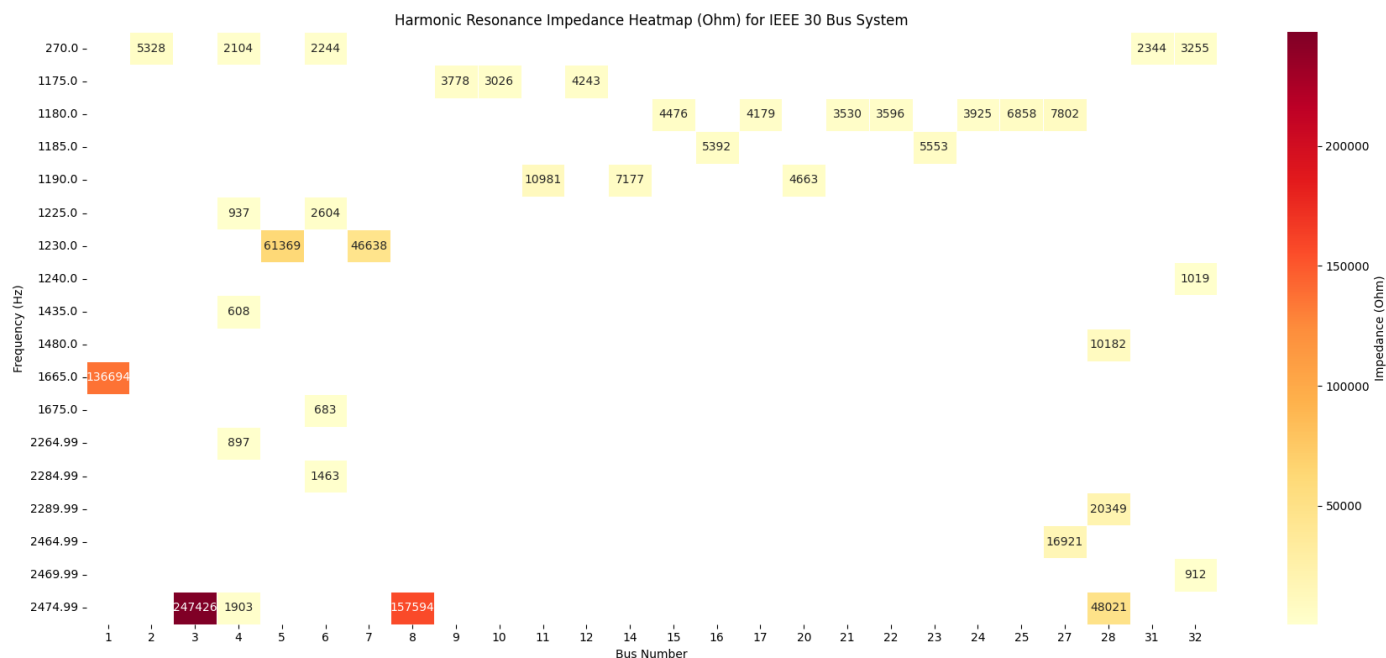


Fig: Harmonic Resonance Impedance Heatmap (Ohm) at the System after Counter Measures Implementation

Following the application of voltage support measures (SVCs and fixed shunts) and the reinforcement of transmission lines, the harmonic environment of the system was reassessed. The harmonic analysis revealed that certain buses, particularly those in close proximity to HVDC converter terminals, exhibited increased levels of harmonic distortion. These distortions were primarily attributed to the firing and commutation behaviors of the converters, which introduced harmonics into the system. The presence of such harmonics can impact system waveform quality, particularly at high-impedance buses where harmonic currents tend to accumulate. The buses whose impedances were the most affected by harmonic frequencies are listed in the following table:

Bus	Harmonic Order	Frequency (Hz)	Impedance (Ohm)
Bus 3	49	2475 Hz	247,425 Ω
Bus 1	33	1665 Hz	136,694 Ω
Bus 8	49	2475 Hz	157,594 Ω
Bus 5	24	1230 Hz	61,368 Ω
Bus 7	24	1230 Hz	46,638 Ω
Bus 28	49	2475 Hz	48,021 Ω

To address these concerns, the following protective measures can be implemented:

- Applying harmonic filters at critical buses, especially those connected to HVDC terminals or where high distortion levels were detected.
- Using damping or filtering mechanisms at HVDC converter buses to reduce the injection of harmonics into the AC network.
- Physically isolating or relocating sensitive equipment (e.g., control systems or precision loads) away from buses with high harmonic impedance or poor waveform quality.

These recommendations aim to reduce the Total Harmonic Distortion (THD) levels and ensure compliance with power quality standards.

8. Discussion

The simulation and analysis of the IEEE 30 Bus system both with and without HVDC integration provided meaningful insights into how HVDC affects the operation of traditional AC networks. This section brings together those findings to reflect on the system behavior, evaluate how technically feasible HVDC integration really is, consider the economic practicality and relate it all to real-world applications.

8.1 Analysis of Simulation Outcomes

The simulations made it clear that while HVDC integration can offer certain advantages, it also introduces a set of challenges. When operating at a low voltage level with a single bridge (like in the 250 kV case) the system handled the HVDC link quite well, with only minor effects on voltage and power flow. But as the HVDC voltage and number of bridges increased, so did the strain on the system. In the 800 kV and 1100 kV scenarios, we started to see more serious issues like undervoltage at remote buses, overloading of key transmission lines and changes in power flow that stressed the network.

Without any corrective action, these effects could compromise system stability. However, when countermeasures were applied such as Static VAR Compensators (SVCs), fixed shunt capacitors, and parallel transmission lines, the situation improved significantly. Bus voltages returned to safe levels and overloaded lines were relieved. That said, harmonic distortion remained a concern even after these measures, showing that while stability was restored, power quality still needed attention.

8.2 Technical Feasibility of HVDC Integration

From a technical perspective, the results suggest that HVDC can be successfully integrated into an AC system like the IEEE 30 Bus network, but only with the right support in place. One of the key findings was that HVDC links behave like constant power loads, which means they don't naturally support voltage levels in the system. As a result, additional voltage regulation and reactive power compensation are essential to keep everything balanced.

That being said, HVDC does have some strong technical benefits. It allows precise control of power flow, makes asynchronous interconnection between regions possible and is more efficient than HVAC over long distances due to lower transmission losses. If the technical challenges like reactive power demand and harmonics are planned for and properly managed, HVDC becomes a solid option for future grid expansion.

8.3 Practical Implications and Real-life Applications

In practice, HVDC has already proven its worth in a variety of real-world scenarios. What our simulation shows is that these real-world projects are right to rely on additional components like SVCs and harmonic filters, which play a crucial role in keeping the system stable and the power clean.

The study also highlights how reinforcing transmission paths such as adding parallel lines isn't just a theoretical solution. It mirrors actual practices used by utility companies when expanding or upgrading power grids. Examples of power companies using multiple parallel lines to stabilize the system are given below:

Location / Project	Voltage Level	Number of Parallel Lines	Purpose
Hoover Dam (USA)	230 kV, 500 kV	3–4	Power evacuation to Las Vegas and California
Agra–Gwalior Corridor (India)	400 kV	3–4	Post-2012 grid upgrade, HVDC support, redundancy
Belo Monte to SE Brazil (Brazil)	±800 kV (HVDC), 500 kV (AC)	2 HVDC + 3 AC	Massive hydro transfer, reliability
China UHV Grid (Xinjiang–Shanghai)	±800 kV, 1000 kV	2–4 UHV lines	Gigawatt-scale power transfer over long distance
New York City Grid (Con Edison)	345 kV, 138 kV	2–5 in some corridors	Urban demand, redundancy, load flexibility

9. Conclusion and Recommendations

9.1 Summary of Key Findings

This study explored the integration of HVDC transmission into the IEEE 30 Bus system and analyzed its effects under different configurations. The simulation results demonstrated that HVDC can significantly alter system behavior, especially as transmission voltage and the number of converter bridges increase. While the base case and low-voltage scenarios remained stable, higher-level HVDC configurations (800 kV and 1100 kV) introduced clear operational challenges including undervoltage conditions, line overloading, and changes in power flow.

However, the study also showed that these issues can be effectively addressed with well-planned countermeasures. The application of Static VAR Compensators (SVCs), fixed shunt capacitors, and additional parallel lines successfully restored voltage stability and reduced transmission corridor stress. Harmonic analysis further highlighted the need for filters at converter terminals to maintain acceptable power quality. Ultimately, the simulations affirmed that HVDC integration is technically feasible when appropriate support systems are implemented.

9.2 Final Recommendations on HVDC Integration

9.2.1 System Stability Perspective

From a system stability standpoint, HVDC integration is only viable at **lower voltage levels**, such as 250 kV, where the impact on bus voltages and power flow remains minimal. As the HVDC transmission voltage increases, the system begins to exhibit signs of instability. These effects become critical at 800 kV and 1100 kV configurations, where the system struggles to maintain voltage levels and line loading within acceptable limits. While various countermeasures can restore stability, their necessity highlights that **the system's inherent capability to support high-voltage HVDC is limited**. Therefore, from a stability perspective, HVDC integration at high voltage levels is **not optimal**, as it introduces considerable complexity and requires substantial support infrastructure to maintain reliable operation.

9.2.2 Economic Perspective

HVDC systems require high initial capital investments due to the cost of converter stations, filters and control infrastructure. These costs are only offset beyond a certain break-even point, which typically occurs at transmission distances of 400 to 700 km for overhead lines, as illustrated in the figure below. In this study, the HVDC line was implemented over a distance of 278 km, which falls well short of the break-even range. As a result, the cost savings from reduced line losses do not compensate for the initial infrastructure investment at this scale. Therefore, from an economic standpoint, integrating HVDC in this scenario is not a viable solution, particularly when compared to conventional HVAC alternatives that are more cost-effective over short to medium distances.

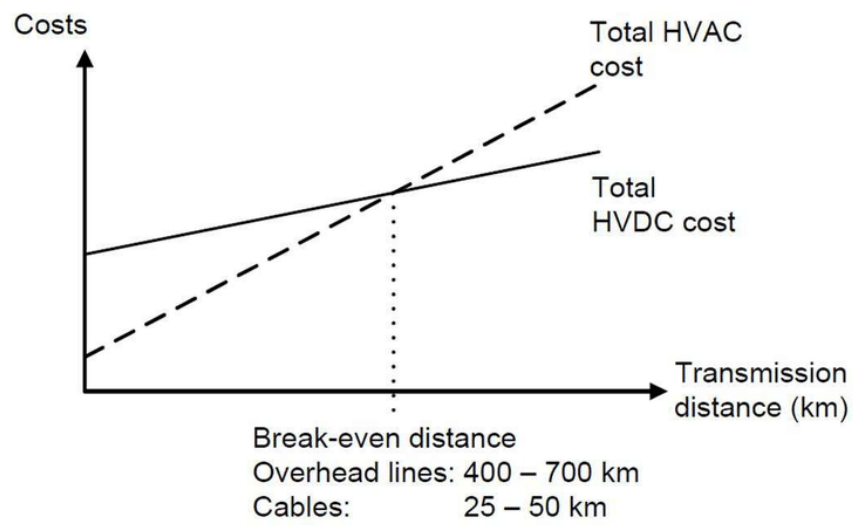


Fig: Cost of HVDC and HVAC system vs Transmission Distance

10. Individual Contributions of Group Members

2106005-Abrar Fahim Niloy	IEEE 30 Bus Drawing and Data Entry,30 Bus Simulation, HVDC Model setup, Case Simulations, Issues Detection, Counter Measures Planning
2106007-Md Abu Saleh Akib	Effect of counter Measures Implementations, Harmonic Analysis in PSSE, Research Paper reference setting, Slide making
2106008-Md Redowanul Hoque	IEEE 30 Bus Drawing and Data Entry,30 Bus Simulation, Counter Measures Planning, Harmonic Analysis Graph simulation code debugging
2106009-Fariha Anjum Oshin	Model Parameters setup, Case Simulations, Issues Detection, Counter Measures Implementations, Harmonic Analysis Graph simulation

11. Outcome-Based Education (OBE) Outcome Achieved

CO No	Description	Corresponding PO No
CO5	Design a power system element with appropriate considerations to safety, cultural, societal, and environmental aspects	PO(c), PO(f), PO(g)
CO6	Demonstrate leadership in designing power system related problem solving	PO(j)
CO7	Communicate effectively on power system element design with a detailed report	PO(i)
CO8	Demonstrate project management and cost analysis for power element design project	PO(k)

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